

Final Project Deliverable

Recent advances in biological data collection have helped in transforming the life sciences. Techniques such as cell RNA sequencing and large-scale cancer genomics projects can now generate massive datasets with thousands of features per sample. While these data provide new biological insights, their size and complexity make analysis extremely challenging using traditional statistical approaches. As a result, machine learning, specifically deep learning, has become increasingly more prominent in the field as a tool for analyzing biological patterns from the data.

This paper synthesizes two recent studies that apply machine learning to these biological dataset problems: Large-Scale Cell Representation Learning via Divide-and-Conquer Contrastive Learning (Zhao et al., 2023) and LASSO-MOGAT: A Multi-Omics Graph Attention Framework for Cancer Classification (Alharbi et al., 2024). Although these studies focus on different biological scales, one on single-cell analysis and the other on patient-level cancer classification, they both share the same objective of learning informative representations from high-dimensional biological data. Together, they help demonstrate how modern machine learning methods can advance biological understanding and medical applications.

A major challenge in biological data analysis is representative learning, which is how raw biological measurements can be transformed into compact summaries that still contain important information. These dense, low-dimensional vectors that represent the original raw biological measurements are called embeddings. For single-cell RNA sequencing, each cell is described by expression levels for tens of thousands of genes, which can result in sparse and noisy data. More traditional deep learning methods, such as BERT-like systems, have been used to model single-cell RNA sequence data. However, these systems often produce poorly separated data that fail to capture meaningful biological variation.

One study on large-scale cell representation learning addresses this problem by proposing a new transformer-based model called CellLM (Zhao et al., 2023). The central hypothesis of this paper is that contrastive learning, where the model learns by comparing similar and dissimilar samples, can produce high-quality cell embeddings. However, a limitation of contrasting learning is that it requires large batch sizes, which can be unrealistic for single-cell RNA sequence datasets because of hardware/technological limitations. In order to overcome this, the authors introduced a novel divide-and-conquer contrasting learning strategy. This strategy allows the model to simulate large batch training by splitting the batches into smaller components, while still allowing for effective contrastive loss.

CellLM was trained through self-supervised learning on about 2 million cells, with each roughly containing 20,000 genes. This model performs tasks like predicting masked gene expression values and determining the differences between healthy and cancerous cells. This allows for learning general patterns without labeled data. Furthermore, CellLM also uses already known biological data through using a gene network encoder, which helps the model understand relationships between genes instead of treating them as independent features. This transformer-based design was chosen because it can scale extremely large datasets and capture complex gene relationships. This makes it well-suited for high-dimensional single-cell data.

While CellLM focuses on learning representations at the level of individual cells, many biological and clinical applications also require understanding data at the level of tissues or patients. Cancer classification, for example, often relies on integrating multiple types of

molecular data (like gene expression, DNA methylation, and genetic mutations. This integration of data is known as multi-omic analysis, and also shows limitations due to the high dimensionality of each layer and the complex interactions.

To solve this, the LASSO-MOGAT (Alharbi et al., 2024) addresses these challenges by combining feature selection with graph-based deep learning. The authors hypothesized that reducing noise through feature selection and clearly modeling biological relationships can help improve cancer classification performance while making it still interpretable. To achieve this, the model uses LASSO, a regularization model that chooses the most informative features and shrinks the weight of the less important features to zero. By doing this, it significantly reduces dimensionality and helps the model focus on the more biologically relevant signals.

After feature selection, the data were organized into a graph structure, where the nodes were the genes and the edges represented biological interactions. A Graph Attention Network (GAT) is trained on this graph and uses attention mechanisms to assign the weights to connections between the nodes. This allows the model to focus on more relevant gene interactions for predicting cancer type. Thus, by integrating multiple omics layers through this system, LASSO-MOGAT produces a final cancer classification while also highlighting which genes and pathways hold the most weight towards the prediction. This model was chosen because graph attention networks naturally represent biological interaction networks and provide interpretability through attention weights, which is critical in cancer research.

Both studies demonstrate a strong performance in their tasks. CellLM manages to achieve much more efficient and biologically relevant results that can be useful in things like drug sensitivity prediction. For example, the model improves macro F1 scores for cell type annotation by approximately 3–5% compared to previous models such as scBERT. Additionally, the model performs well in settings where only a small amount of labeled data is available. Similarly, LASSO-MOGAT showed high classification accuracy across 31 cancer types (using data from The Cancer Genome Atlas). This model reaches classification accuracies above 90% for many cancer types and consistently outperforms traditional machine learning methods such as support vector machines and random forests. One strength of the LASSO-MOGAT is its robustness for missing data, which can be a common issue in real-world clinical datasets.

When compared together, these studies show similar strategies for handling biological complexity. CellLM emphasizes large-scale, self-supervised learning to allow to model to generalize across diverse cell types and tasks. Meanwhile, LASSO-MOGAT emphasizes learning and interpretability through using biological graphs to classify types of cancer. Both of these studies show a trend in life sciences toward combining biological knowledge with advanced machine learning techniques. Hence, these studies demonstrate that machine learning can now do more than improve predictive accuracy. It can also help in improving biological understanding.

With the rapid growth of biological data, machine learning is slowly becoming a more essential tool in modern life sciences. The two studies here both illustrate how deep learning methods can help address challenges in both single-cell analysis and multi-omics cancer classification. Through contrastive learning and graph attention networks, these models can help extract meaningful patterns from high-dimensional data to help in both research and clinical applications. As machine learning continues to evolve, it is clear that the integration of machine learning to make data more interpretable will play a critical role in biological discovery and medicine.

References

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